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SECP3133 - 01 HIGH PERFORMANCE DATA PROCESSING

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ASSIGNMENT 1

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UNDERSTANDING THE REAL-TIME QUERY EXECUTION OF APACHE DRILL

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Abstract

As data velocity increases via Internet of Things (IoT) and intelligence device, Apache Drill offers a high-performance solution for real-time ad-hoc analytics. This paper examine Drill architecture and its drillbits execution model which minimize latency through data locality. Comparative analysis shows that Drill approach outperforms Apache Hive in scalability and stability. The results shows that at a 100 GB scale, Drill speed up is nearly double the Hive making it superior for real time data exploration.

Keywords Apache Drill, Real time analytics, Apache Hive, Scalable SQL system

Introduction

With the advancement in intelligent devices, social media, and the Internet of Things, the amount of new data being generated is high, and the pace is continuously accelerating. Real-time analytics (RTA) has emerged as a distinct branch of big data analytics focusing on the velocity aspect of big data, in which data is prepared, processed, and analyzed as it arrives, intending to generate insights and create business value in near real-time. Apache drill is one of the platforms or tools for real-time analytics as it is designed to handle up to petabytes of data spread across thousands of servers, and the goal of Drill is to respond to ad-hoc queries in a low latency manner. This paper discusses the use of Apache Drill within real-time analytics frameworks. It provides detailed examination of the Apache Drill high-level architecture and presents a comparative analysis with Apache Hive, which is another scalable SQL system to evaluate performance, schema flexibility, and scalability.

Technology Explanation

As shown in Figure 1, Layer 3 is the key area of this paper, which is the real-time analytics platform. In this layer, it interprets the data from the second layer to detect patterns, trigger alerts, execute actions or create new events. Typically, layer 3 platforms have few core parts which are event metadata stores event types and processing rules . Event Processing Language (EPL) is used to describe event patterns, and EPL compilers turns those information into code so that the engine can execute. The Event Processing Engine matches the pattern types against incoming data and trigger the alerts or actions. For example, in the case study of Bakker et al. (2020), it mentioned that the real time analytics act as a clinical decision support tool for identifying ventilator patients that mismatch in real time. With real time analytics platform, it could detect ineffective efforts events and compare current care with analytics assisted care for mechanically ventilated ICU patients. The

outcomes of this analytics is such as hospital mortality, ICU length of stay, and life years.

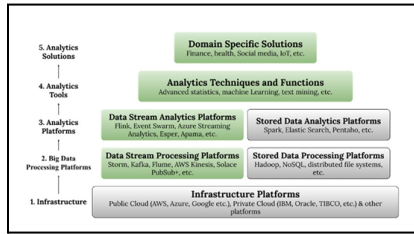


Figure 1 Real time analytics stack

The technology use in this real-time analytics platform is Apache Drill or better known as a distributed, interactive ad hoc analytics query engine. A query is usually entered through Java Database Connectivity (JDBC) or Open Database Connectivity (ODBC) as they are client interfaces that allow BI apps or SQL clients to send queries to Drill. As the data comes from JDBC or ODBC, Drill parses it into a logical plan where it executes strategy, including parallelism and use of columnar data when possible. Then, transformed and optimized it into a physical plan. Execution is distributed among workers and called as drillbits, which run on processing nodes to keep computation close to the data. From different sources, Drill uses storage connectors and scanners to read and it can push parts of the query down into the source system so that it can reduce work and speed things up.

Architecture Discussion

As shown in Figure 2 is the Apache Drill high level architecture. This architecture is the blueprint of the Apache Drill workflow from client to execution. This architecture works by a user or tool sends SQL through JDBC or ODBC. Then it parse the SQL into a logical plan. The logical plan is then reshaped into a physical plan by using cluster topology and parallelism. Moreover, the physical plan is distributed by drillbits across nodes to run each processing

node to maximize data locality. Those drillbits manage the coordination, scheduling, and execution so that the computation happens close to where the data live. Furthermore, Drill is flexible for mixed data sources and helps performance such as exploratory analysis over large dataset. This is because Drill connect to multiple storage system through scanners and push parts of the query down into the source when possible.

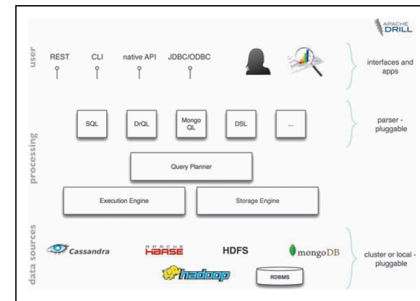


Figure 2 Apache Drill high level architecture

Therefore, query layer in the middle of the stack is the function of Drill's core. It sits in between clients and data sources to handle parsing, planning, optimization and distributed execution rather than stores the data itself. Overall, Apache Drill plan and distribute the work from the query that the client sends, then drillbits execute it and the underlying sources provide data.

Comparative analysis

Apache Hive is taken into consideration to make analysis with Apache Drill because both of them are SQL scalable systems. Based on the analysis made by Cardas et al. (2022) and Mark-Andor (2025), it shows that Apache Drill and Apache Hive have a huge difference in terms of schema, governance, performance with 100GB and scalability analysis.

Table 1 shows the comparative analysis of Apache Drill and Apache Hive:

Criteria	Apache Drill	Apache Hive
Schema	-schema-on-read -discovers the schema at query time	-schema-on-write -predefined schema
Governance	Ad hoc and exploratory	Stronger fit for a controlled data warehouse environment
Performance with 100 GB scale factor	-Execution time mostly stable with low dispersion in most queries -Average speed-up is 2.21	-Larger variation on some queries -Average speed-up is 1.09
Scalability analysis	Improved with scale and became the second fastest on several queries	Degrade further as data size increases

Table 1 Comparative analysis

The reason of this difference are:

- *Schema*

In Apache Drill, it query files directly without predefining a table schema. It discovers schema at query time, which is good for semi-structured, or changing data format like JSON, CSV, and Parquet. Meanwhile, Apache Hive is a predefined schema in metastore, and data is expected to fit the structure when it's loaded or queried. This is beneficial for curated, warehouse-style dataset. (Mark-Andor, 2025)

- *Governance*

Hive is more stronger for a controlled data warehouse environment because the schema is defined up front in the metastore. Teams could standardize table definitions, enforce naming conventions, manage partitions and apply access control and auditing using known datasets (Mark-Andor, 2025). Meanwhile, as mentioned by Rodríguez-Mazahua et al. (2026) Drill is

more ad hoc and exploratory because it can query files directly and change structure during runtime. It's easy to analyze data quickly but at the same time the governance is harder because of less central control over schema definitions, certified dataset and data quality rules. (Mark-Andor, 2025)

- *Performance with 100 GB scale factor*

Apache Drill is more stable because it uses HDFS views and schema inference, so the behaviour stays fairly consistent meanwhile Apache Hive struggles as larger variation because the shuffling creates I/O bottlenecks when data grows. (Cardas et al., 2022)

- *Scalability analysis*

This analysis was made to examine how those same behaviors change as data grows from 10 GB to 100 GB to 300 GB. The result is Drill scales better and more smoothly than Hive in this study. This is because when queries involve large tables, joins, or nested queries, Hive suffers from I/O bottlenecks during shuffling, so its performance worsens more sharply as the scale factor increases. Drill, by contrast, stays more consistent because it can infer schema from HDFS data and use views, which helps keep execution behavior steadier across queries. (Cardas et al., 2022)

Conclusion

In conclusion, Apache Drill serves as a powerful engine for the real time insights as it offers a flexible query layer that bridges the gap between complex and end-user business intelligence tools. With its unique architecture that make use of logical and physical plans distributed across drillbits, ensures that computation remain close to the data, significantly reduce latency and maximizing throughput for ad-hoc queries. The comparative analysis also has highlighted Drill's

specific strength in agility and performance. Overall, technology like Apache Drill will remain longer for organizations to turn raw data, real time streams into actionable outcomes.

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Logbook Assignment 1

Date of works	Search keywords	Databases used	AI prompts used	Task description
23/4/2026	1. Real-Time Analytics Platforms 2. Apache Drill	- Scopus - Web of science	- How real time analytics works - What is Apache Drill	- Find related paper - Extract the highlighted sentences
24/4/2026	1. Real-Time Analytics Platforms 2. Apache Drill Architecture	- Scopus - Web of science	- In which layer drill at - How this drill architecture works	- Extract and rephrase the text from the article
25/4/2026	1. Apache Drill 2. Apache Drill Architecture 3. Apache Hive	- Scopus - Web of science	- What is the significant difference between Apache Drill and Apache Hive	- Make comparison analysis between drill and hive in terms of schema and governance - Analysis criteria that should be compared
26/4/2026	1. Apache Drill 2. Apache Drill Architecture 3. Apache Hive	- Scopus - Web of science	- What influence hive and drill performance and scalability	- Find article and understand to support the reason the drill is better than hive - Compile all the findings and write in one article
27/4/2026				- Double check the article - Upload in turnitin